Prototype-Conditioned Generative Adversarial Network for Few-Shot Remote Sensing Object Detection

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***Abstract*—Few-shot object detection (FSOD) in remote sensing imagery faces critical challenges stemming from extreme data scarcity, specifically inadequate feature coverage, severe class imbalance, and pervasive incomplete annotations. To address these interconnected issues, this article proposes a unified FSOD framework based on a prototype-conditioned GAN (P-GAN). The framework integrates three components to enhance robustness from data, feature, and label perspectives. First, dynamic augmented balanced sampling (DABS) is introduced to mitigate overfitting by applying diverse, adaptive augmentations to oversampled novel-class instances, improving both numerical balance and visual diversity. Second, to alleviate feature scarcity, a method is designed to synthesize region-of-interest features. Guided by prototype vectors and a cross-attention mechanism, P-GAN helps expand the feature space and alleviate classifier bias toward base classes. Third, to tackle the false negatives caused by missing annotations, a dynamic prototype-aware label corrector (DPLC) exploits a teacher–student architecture and prototype similarity to adaptively recalibrate labels and loss weights. Experiments on the DIOR and NWPU VHR-10.v2 benchmarks show that the proposed approach improves detection performance over the compared methods across multiple shot settings by mitigating classifier bias and refining feature representations.**

***Index Terms*—Few-shot learning, generative adversarial networks (GANs), object detection, prototype learning.**

I. Introduction

OBJECT detection is a fundamental task in computer vision, with extensive applications in remote sensing image analysis [1], [2]. Recent advances in deep learning have improved object-detection frameworks [3], from early region-based methods [4], [5] to efficient single-stage detectors [6] and Transformer-based architectures [7]. However, these detectors generally depend on large-scale annotated training datasets. Acquiring such annotations for remote sensing imagery is expensive and can be impractical. Consequently, few-shot object detection (FSOD) has emerged as a research direction for settings with severe data scarcity [8], [9].

Following research on few-shot image classification, FSOD can be grouped into two primary training paradigms: meta-learning-based approaches [9], [10] and transfer-learning-based approaches [11]. Meta-learning-based methods train models on episodic tasks that simulate few-shot conditions [12], [13]. In contrast, transfer-learning-based approaches generally comprise base training and fine-tuning. During base training, the model learns general feature representations from a base dataset with abundant annotations. During fine-tuning, it adapts to novel categories using a small number of annotated samples [14], [15]. Despite recent progress, obtaining reliable detection accuracy from a limited number of examples remains challenging.

The efficacy of FSOD in remote sensing is constrained by two deeply interconnected challenges: extreme feature scarcity and pervasive incomplete annotations. Prior works often treat these as isolated issues, relying on either generative feature augmentation [16] to alleviate scarcity or label correction techniques [17], [18] to address missing labels. However, in complex remote sensing scenarios featuring dense and ambiguous objects, these two challenges can form a harmful cycle. Data scarcity inherently limits feature coverage, making models highly sensitive to the available support samples. Simultaneously, incomplete annotations [19] cause unlabeled foreground objects to be erroneously assigned as background, introducing severe false negatives [20]. Under few-shot conditions, these mislabeled regions propagate noisy gradients that may distort the already fragile category prototypes. Consequently, corrupted labels can lead to biased prototypes, which in turn cause generative modules to synthesize noisy, uninformative features, ultimately creating a circular dependency that exacerbates classifier shift toward base classes [21], [22].

To address these challenges in remote sensing imagery, we propose a unified FSOD framework based on Faster R-CNN that combines a prototype-conditioned generative adversarial network (P-GAN) with a dynamic prototype-aware label corrector (DPLC). The two modules share a dynamic prototype queue. Within the region of interest (ROI) head, DPLC modulates the loss weights of ambiguous negative proposals according to their cosine similarity to class prototypes, reducing false-negative contamination through a teacher–student architecture. P-GAN uses the refined category-specific prototypes to synthesize ROI features for novel classes. This coupling allows feature synthesis and label refinement to use the same prototype information, helping to rebalance class distributions and alleviate classifier bias.

Furthermore, to alleviate overfitting during fine-tuning caused by the simple oversampling of novel-class instances, we introduce dynamic augmented balanced sampling (DABS). Rather than naively repeating samples, DABS applies adaptive,

Received 12 February 2026; revised 8 May 2026 and 23 July 2026; accepted 31 August 2026. Date of publication 3 September 2026; date of current version 14 September 2026. This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62301444. *(Corresponding author: Dandan Ma.)*

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Digital Object Identifier 10.1109/TGRS.2026.3730249

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diverse data augmentation techniques to duplicated novel-class images. This strategy maintains numerical interclass balance while increasing visual diversity, which can improve robustness during the early training stages.

To address these interconnected challenges, we propose a unified FSOD framework built upon three components. The framework targets feature scarcity through prototype-conditioned feature synthesis, annotation incompleteness through adaptive label refinement, and oversampling-related overfitting through dynamic data balancing. The key contributions of this work are as follows.

1) To tackle the cycle of feature scarcity and false-negative contamination, we propose a synergistic joint design comprising P-GAN and DPLC. The contribution of this design is that by explicitly coupling feature synthesis and label refinement through a shared dynamic prototype space, it helps alleviate the limitations of traditional isolated FSOD pipelines and supports the co-evolution of features and labels.
2) To address the low intraclass diversity caused by novel-class data scarcity, we design the P-GAN module. Conditioned on dynamically updated prototype vectors, P-GAN synthesizes varied ROI features, helping to expand the target feature space and alleviate classifier shift.
3) To address class confusion induced by incomplete annotations in remote sensing datasets, we introduce the DPLC module. By leveraging a teacher–student architecture and prototype similarity, DPLC adaptively recalibrates the labels and loss contributions of ambiguous background proposals. This mechanism reduces the negative influence of false negatives and improves the stability of the shared prototype space.
4) To mitigate overfitting caused by oversampling novel-class instances, we introduce DABS. Instead of naively repeating samples, DABS applies a dynamic and diverse set of augmentations to each oversampled instance. This strategy maintains numerical balance between base and novel classes while increasing visual diversity, thereby reducing repeated exposure to identical few-shot samples.

II. Related Work

A. Few-Shot Object Detection

FSOD aims to recognize novel categories with minimal labeled examples, primarily through two paradigms: meta-learning and transfer learning.

Meta-learning methods simulate few-shot scenarios via episodic training to acquire task-level knowledge. For instance, Meta R-CNN [9] extracts class-specific attention vectors from support samples to reweight ROI features, intrinsically prioritizing discriminative information. To further mitigate overfitting under data scarcity, Han et al. [17] utilize variational autoencoders for probabilistic feature modeling, while Karlinsky et al. [23] jointly optimize feature representations alongside distance metrics to enhance generalization.

Despite their efficacy, meta-learning approaches often require complex training pipelines. In contrast, transfer learning offers a more streamlined and highly effective alternative. TFA [24] established a standard two-stage fine-tuning paradigm, efficiently transferring prior knowledge from abundant base classes to novel ones. Building upon this, Sun et al. [15] introduced contrastive proposal encoding to maximize intraclass similarity and interclass variance, significantly reducing category confusion. Recently, Wang and Yu [25] proposed an orthogonal progressive model that cleanly decouples classification and regression tasks, utilizing a progressive fine-tuning strategy with historical features to effectively alleviate overfitting.

B. FSOD in Remote Sensing

Recent advances in FSOD for remote sensing have yielded various strategies to mitigate category diversity and sample scarcity. Metric learning approaches frequently leverage representative features; for instance, Cheng et al. [14] employ a prototype convolutional neural network with contrastive loss to enhance interclass separability, while Yang et al. [26] introduce a prototype-based meta-prompt tuning strategy to enable rehearsal-free class-incremental learning in multimodal imagery. To capture complex scene contexts, Zhang et al. [27] utilize graph neural networks to model object relationships. Recent remote sensing detection studies have also emphasized adaptive refinement and context-aware modeling under complex imaging conditions. Feng et al. [28] proposed a coarse-to-fine dynamic refinement model for weakly supervised moving vehicle detection in satellite videos that refines detections progressively. Feng et al. [29] also introduced a semantic-embedded density-adaptive network for moving vehicle detection in satellite videos that incorporates semantic embedding and density-adaptive modeling for dense scenes. These studies further motivate adaptive refinement and semantic-aware modeling for robust remote sensing detection. Alternatively, architectural adaptations such as dual-head structures have proven effective. Wolf et al. [30] modulate feature representations using condition vectors to jointly optimize classification and regression, and Liu et al. [20] design a label-consistent dual-head classifier that leverages base-class priors to correct novel-class label inconsistencies under extreme data scarcity.

C. Generative Data Augmentation for Object Detection

Generative adversarial networks (GANs) effectively mitigate data scarcity in FSOD by dynamically enriching the feature space [31]. Specifically, perceptual GAN leverages a scale-aware transformer to project shallow features into discriminative super-resolution representations, generating multiscale variants to ensure comprehensive feature coverage for novel classes. To combat classification bias induced by long-tailed distributions, Wang et al. [32] proposed a dynamic feature alignment module utilizing a class-balanced adversarial loss to match base and novel feature distributions. Moreover, Lin et al. [33] developed a multitask boundary-aware discriminator that jointly assesses feature authenticity

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and optimizes class boundaries. Extending this generative capability to complex domains, Zhang et al. [34] introduced a physics-inspired Φ-GAN to synthesize high-fidelity SAR images under limited data regimes. Collectively, these studies indicate the usefulness of GAN-driven feature augmentation for addressing FSOD challenges.

D. Incomplete Annotation

In FSOD, incomplete annotations generate false negatives that severely misguide the learning process. Current mitigation strategies primarily target region proposal network (RPN) sampling and ROI classification through pseudo-label supplementation or bias suppression. For instance, Zhang et al. [18] utilize a teacher–student self-training framework to mine high-quality proposals and correct erroneous high-confidence background labels. To maintain label consistency, Liu et al. [20] introduce a dual-head detector during fine-tuning, which refines novel-class predictions using background probabilities derived from the base-class classifier. Alternatively, Yan et al. [22] isolate ambiguous samples using a CPS algorithm and apply adaptive gradient regulation, dynamically modulating the gradient optimization coefficients based on prototype-ROI similarity to effectively alleviate foreground-background confusion.

III. Methodology

In this section, we first introduce the problem definition and the training paradigm to clarify our task. Subsequently, we provide an overview of our method as shown in Fig. 1, followed by detailed descriptions of the DABS, DPLC, and P-GAN modules, which address the challenges of class imbalance and incomplete annotations in FSOD, respectively.

A. Problem Definition and Training Paradigm

1) Problem Definition: Following previous work [35], we adopt the standard FSOD formulation. The category set is divided into disjoint base and novel subsets, denoted by $\mathcal{C}_{base}$ and $\mathcal{C}_{novel}$, respectively, where $\mathcal{C}_{base} \cap \mathcal{C}_{novel} = \emptyset$. The training set $\mathcal{D}_{train} = \{(I, B)\}$ comprises a base set $\mathcal{D}_{base} = \{(I_i, B_i^{base})\}$ with abundant annotations and a novel set $\mathcal{D}_{novel} = \{(I_i, B_i^{novel})\}$ with limited annotations. For each sample $(I_i, B_i) \in \mathcal{D}_{train}$, I_i denotes an image and $B_i = \{(x, y, w, h, c)\}$ denotes its annotated bounding boxes, where (x, y, w, h) parameterizes a bounding box and c is the class label.

2) Training Paradigm: Our method follows the standard two-stage TFA paradigm [24]. In the base-training stage, the detector is trained on $\mathcal{D}_{base}$ to initialize the backbone, RPN, and ROI head. In the fine-tuning stage, the detector is initialized with the base-training weights and fine-tuned on $\mathcal{D}_{ft} = \{(I_i, B_i^{base}, B_i^{novel})\}$, which contains K annotated instances for each novel class together with base-class instances. For each novel class $c \in \mathcal{C}_{novel}$, we have

$$\sum_{(I_i, B_i^{base}, B_i^{novel}) \in \mathcal{D}_{ft}} \left| \left\{ (x, y, w, h, c) \in B_i^{novel} \right\} \right| = K. \tag{1}$$

Here, $|\cdot|$ indicates the number of labeled samples of each of the novel classes.

Algorithm 1 Dynamic Augmented Balanced Sampling

Input Training image set $\mathcal{D} = \mathcal{D}_{base} \cup \mathcal{D}_{novel}$
Output Balanced fine-tuning set $\mathcal{D}_{ft}$

$R = \lceil \frac{|\mathcal{D}_{base}|}{|\mathcal{D}_{novel}|} \rceil$
$\mathcal{D}_{ft} = \mathcal{D}_{base} \cup \underbrace{\mathcal{D}_{novel} \cup, \ldots, \cup \mathcal{D}_{novel}}_{R \text{ times}}$
Assign a repeat index t_i for each image I_i in $\mathcal{D}_{ft}$
Calculate the small-object indicator b_i for each image I_i in $\mathcal{D}_{ft}$
if $t_i = 0$ **then**
    Apply basic augmentation
else
    if $b_i = 1$ **then**
        Apply small object augmentation variant
    else
        $v = t_i \mod 3$
        Apply the v-th augmentation variant
        if $v = 0$ **then**
            Apply color jitter augmentation
        else if $v = 1$ **then**
            Apply grid occlusion augmentation
        else if $v = 2$ **then**
            Apply small object duplication augmentation
        end if
    end if
end if

B. Dynamic Augmented Balanced Sampling

In FSOD, constructing the fine-tuning set $\mathcal{D}_{ft}$ requires balancing the sample distributions of base and novel classes. Standard approaches such as TFA use fixed-K sampling to maintain interclass balance. However, straightforward oversampling of the limited novel-class images can cause overfitting because the same visual content is repeatedly observed. It also provides limited augmentation for the small objects that are prevalent in remote sensing imagery. Wolf et al. [30] showed that increasing the number of base-class samples and oversampling novel-class objects can improve detection performance, but the repeated samples still require sufficient visual diversity.

To address this issue, we propose DABS. Each oversampled novel-class image is assigned a repeat index, and DABS selects an augmentation according to that index. Original instances receive basic augmentation; duplicated images with small objects receive a dedicated small-object augmentation, while the remaining duplicated images cycle through color jitter, grid occlusion, and small-object duplication. This strategy preserves numerical class balance while increasing the visual diversity of repeated novel-class samples. The procedure is summarized in Algorithm 1.

C. Dynamic Prototype-Aware Label Corrector

Incomplete annotations in remote sensing FSOD can generate noisy gradients by assigning unlabeled foreground objects as background. To mitigate this issue, we propose a DPLC that refines label assignments within the ROI head. DPLC

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relies on a dynamic prototype queue [36], [37], a structured memory bank that maintains evolving class-specific feature distributions for label calibration.

The dynamic prototype queue uses an adapter network with first-in-first-out and momentum updates to maintain a low-dimensional ROI feature space and category prototypes [38]. A dynamic prototype contrastive loss optimizes the adapter outputs by pulling features toward their respective prototypes and pushing distinct categories apart [39]. This encourages separable representations for downstream detection [40]

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\left[c_i \in \mathcal{C}\right]\cdot\log\frac{\exp\left(s\left(\mathbf{f}_i,\mathbf{p}^{(c_i)}\right)\right)}{\sum_{c'\in\mathcal{C}}\exp\left(s\left(\mathbf{f}_i,\mathbf{p}^{(c')}\right)\right)} \tag{2}$$

where N is the number of ROI features in the batch, c_i is the class label of $\mathbf{f}_i$, $\mathbb{I}[\cdot]$ is the indicator function, $\mathbf{p}^{(c)}$ is the prototype vector of class c, and $s(\cdot,\cdot)$ denotes cosine similarity. To prevent prototype redundancy and encourage diversity in the embedding space, we incorporate a regularization term that penalizes high cosine similarity between distinct prototypes [41]

$$\mathcal{L}_{\text{div}} = \frac{1}{|\mathcal{C}|\left(|\mathcal{C}|-1\right)}\sum_{j\neq k}\exp\left(s\left(\mathbf{p}^{(c_j)},\mathbf{p}^{(c_k)}\right)\right). \tag{3}$$

The final loss combines both objectives with a balancing coefficient $\mu = 0.1$

$$\mathcal{L}_{\text{APCL}} = \mathcal{L}_{\text{contrast}} + \mu\cdot\mathcal{L}_{\text{div}}. \tag{4}$$

Leveraging the prototype vectors, the DPLC module adaptively calibrates labels for negative proposals that fall within the gray area of teacher-network confidence. Let $u^{stu}\in\mathbb{R}^{|\mathcal{C}|+1}$ denote the class probability distribution predicted by the student network, and let $\hat{y}^{tch}$ denote the pseudo-label predicted by the teacher network. The base classification loss function L_{cls} is defined as the standard cross-entropy loss

$$L_{\text{cls}}\left(u,y\right) = -\log\left(u_y\right) \tag{5}$$

where u_y is the predicted probability for the target class y.

For the ith ambiguous negative proposal, we calculate its maximum cosine similarity to all class prototypes, denoted as $s_i = \max_{c\in\mathcal{C}}\cos(\mathbf{f}_i,\mathbf{p}^{(c)})$. A high similarity score indicates that the proposal is likely a missed foreground object. Therefore, we transform this similarity score into a prototype-aware modulation weight w_i using a continuous mapping function

$$w_i = \alpha + (1-\alpha)\exp\left(-\exp\left(-\beta\left(1-s_i\right)\right)\right) \tag{6}$$

where α and β are hyperparameters that govern the baseline weight and the steepness of the exponential transition, empirically set to 0.25 and 20, respectively.

The classification loss for negative proposals is dynamically modulated by the teacher prediction. When the teacher assigns a foreground pseudo-label, the student is supervised by that pseudo-label. Otherwise, the background loss is attenuated by the prototype-aware weight w_i. This adaptive strategy suppresses detrimental gradients induced by potential false negatives

$$\mathcal{L}_{\text{cls}}^{\text{neg}} = \begin{cases} L_{\text{cls}}\left(u_i^{stu},\hat{y}_i^{tch}\right), & \hat{y}_i^{tch}\neq 0 \\ w_i\cdot L_{\text{cls}}\left(u_i^{stu},0\right), & \hat{y}_i^{tch}=0. \end{cases} \tag{7}$$

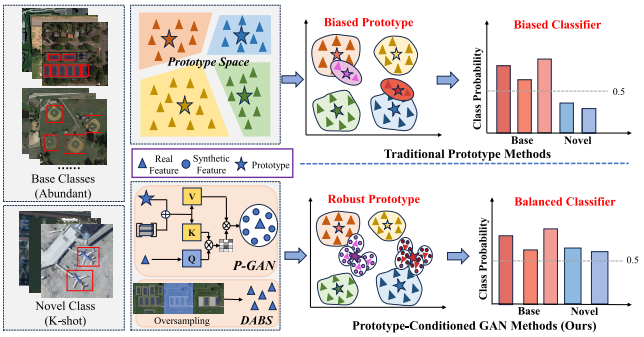


Fig. 1. Overview of the proposed framework for few-shot remote sensing object detection. The method constructs prototypes from base-class samples and uses P-GAN to synthesize features for novel classes with limited data. DABS and DPLC support balanced training and label correction, respectively.

This mechanism reduces the influence of potential false negatives during few-shot fine-tuning.

D. Prototype-Conditioned GAN

Building on the feature representations maintained by the dynamic prototype queue, we introduce the P-GAN, an adversarial module for alleviating novel-class feature scarcity [31]. P-GAN operates at the ROI level to synthesize features during training, augmenting the feature space of novel classes within each mini-batch. The architecture comprises a conditional generator [42] and a multiscale discriminator [43], as shown in Fig. 2.

The conditional generator takes random noise and, more importantly, a conditioning signal derived from the class prototype vector of a target novel class as input. To achieve more precise semantic alignment and feature synthesis, we incorporate a cross-attention mechanism within the generator. In this mechanism, the class prototype and the noise vector are concatenated and projected to serve as the key and value. Simultaneously, a randomly selected real ROI feature from the same target class is used as the query [44]. This allows the generator to attend to the most relevant semantic aspects of the class prototype while maintaining the structural characteristics of real ROI features.

This direct, attention-based guidance encourages the generated features to be semantically anchored to the target class while exhibiting realistic variations. The generator produces plausible ROI feature embeddings that better approximate the real features of that class stored in the prototype queue. The generated features are then introduced into the training pipeline to augment the real features of the novel class. To enhance their discriminative quality, we apply a cross-entropy loss by feeding them to the classifier

$$\mathcal{L}_{\text{cls}}^{\text{gen}} = -\mathbb{E}_{z\sim p_z(z),c\sim p_{novel}}\left[\log P\left(y=c\middle|G\left(z,\mathbf{p}_c\right)\right)\right] \tag{8}$$

where z is the noise vector, c is the novel class label, $\mathbf{p}_c$ is the prototype for class c, and $G(z,\mathbf{p}_c)$ is the generated feature. This loss encourages the synthesized features to be correctly classified by the detector's classifier, thereby contributing to the mitigation of classifier bias.

The multiscale discriminator distinguishes real features from synthesized features. Its multiscale design evaluates

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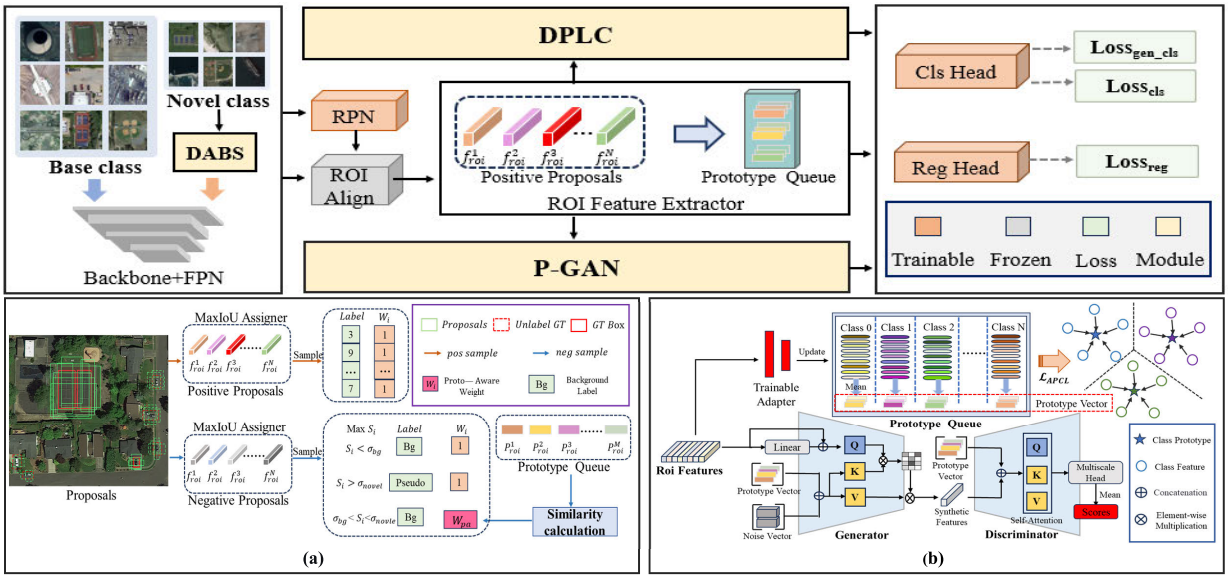


Fig. 2. Overarching structure of the proposed method. The upper section presents the overall detection pipeline incorporating the DABS, DPLC, and P-GAN modules. The lower section details the specific architectures of two core submodules: (a) DPLC for mitigating missing annotations and (b) P-GAN for feature space augmentation.

generated features at different levels of abstraction. The discriminator is optimized in a separate adversarial branch, whereas the generated features contribute to the detector-side optimization through the generated-feature classification loss defined above. As a training-stage module, P-GAN enriches the feature space for novel classes without altering the inference architecture of the detector.

The discriminator adversarial loss distinguishes real and synthesized features [45]. Here, $D(\mathbf{f}, \mathbf{p}^{(c)})$ denotes the discriminator output obtained by averaging the outputs from the three discriminator heads

$$\mathcal{L}_{\text{real}} = \mathbb{E}\left[\log\left(1+\exp\left(-D\left(\mathbf{f}_{\text{real}}, \mathbf{p}^{(c)}\right)\right)\right)\right] \tag{9}$$

$$\mathcal{L}_{\text{fake}} = \mathbb{E}\left[\log\left(1+\exp\left(D\left(\mathbf{f}_{\text{syn}}, \mathbf{p}^{(c)}\right)\right)\right)\right]. \tag{10}$$

The discriminator is optimized in a separate adversarial branch. Its objective is

$$\mathcal{L}_{\text{dis}} = \frac{1}{2}\left(\mathcal{L}_{\text{real}} + \mathcal{L}_{\text{fake}}\right). \tag{11}$$

To avoid ambiguity, $\mathcal{L}_{\text{dis}}$ is used only to update the discriminator parameters and is not directly added to the detector objective. The generator receives gradients through the generated-feature classification loss. During fine-tuning, the generator, detector, DPLC module, and prototype adapter are optimized by minimizing the following multitask objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{RPN}} + \mathcal{L}_{\text{ROI}}^{\text{DPLC}} + \lambda_{\text{APCL}}\mathcal{L}_{\text{APCL}} + \mathcal{L}_{\text{cls}}^{\text{gen}} \tag{12}$$

where $\mathcal{L}_{\text{ROI}}^{\text{DPLC}}$ denotes the combined classification and regression losses in the ROI head dynamically weighted by the DPLC module; $\mathcal{L}_{\text{APCL}}$ is the adaptive prototype contrastive loss (APCL), whose internal diversity term is weighted by μ; $\mathcal{L}_{\text{cls}}^{\text{gen}}$ represents the generated-feature classification loss; and λ_{APCL} is the balancing hyperparameter for APCL in the detector objective. In practice, each training iteration alternates between updating the discriminator with $\mathcal{L}_{\text{dis}}$ and updating the generator and detector-side components with $\mathcal{L}_{\text{total}}$.

IV. Experimental Results

In this section, we conduct experiments on two widely adopted remote sensing datasets to evaluate the performance of our proposed model. To substantiate its effectiveness, comprehensive comparisons are made against both general few-shot object detectors and specialized remote-sensing few-shot detectors. Furthermore, ablation studies are performed to dissect the contributions of key components to the overall detection efficacy.

A. Datasets

1) DIOR: DIOR is a large-scale benchmark dataset for object detection in optical remote sensing images [46]. It comprises 23 463 images and 190 288 annotated instances spanning 20 categories. All images have a uniform width of 800 pixels and a spatial resolution ranging from 0.5 to 30 m. The dataset is divided into training, validation, and test sets containing 5682, 5863, and 11 738 images, respectively.

2) NWPU VHR-10.V2: NWPU VHR-10.v2 [47] is a benchmark dataset curated by Northwestern Polytechnical University for object detection in very high-resolution remote sensing imagery. It comprises 1172 images collected from Google Earth and Vaihingen data and resized to 400 $\times$ 400 pixels. Following common practice, we use NWPU VHR-10.v2 to assess the proposed detector's performance.

B. Experimental Settings

To evaluate the detection performance of our model under the FSOD setting, we divide each dataset into base and novel classes. Each base class has abundant annotations, whereas each novel class has only K labeled instances. We evaluate $K \in \{3, 5, 10, 20\}$ shots per class using annotations from the training and validation sets.

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TABLE I
BASE–NOVEL CATEGORIES SPLIT SETTING

Dataset	Split	Novel Categories					Base Categories
DIOR	1	airplane	baseball field	tennis court	train station	windmill	rest
	2	baseball field	basketball court	bridge	chimney	ship	rest
	3	airplane	airport	expressway toll station	harbor	ground track field	rest
	4	dam	golf course	storage tank	tennis court	vehicle	rest
	5	expressway service area	overpass	stadium	train station	windmill	rest
NWPU VHR-10.v2	/	airplane		baseball diamond		tennis court	rest

For DIOR, we adopt five established base/novel partitioning strategies, as detailed in Table I. The first split, proposed by Li et al. [13], designates airplane, baseball field, tennis court, train station, and windmill as novel classes; it uses the training set for optimization and the validation set for evaluation. The subsequent four splits, introduced by Cheng et al. [14], each contain 15 base and 5 novel classes. For these splits, models are optimized on the combined training and validation sets and evaluated on the test set. For NWPU VHR-10.v2, airplane, baseball diamond, and tennis court serve as novel classes, and the remaining categories serve as base classes. The model is fine-tuned on the combined training and validation sets, with final performance reported on the 293 test images.

Following the PASCAL VOC 2007 benchmark, we use mean average precision (mAP) at an IoU threshold of 0.5 as the evaluation metric.

C. Implementation Details

The proposed method uses Faster R-CNN with a ResNet-101 [48] backbone pretrained on ImageNet. An FPN [49] produces multiscale features. The detector is optimized with AdamW [50], a weight decay of 0.01, and an initial learning rate of 10^{-4}. Base training lasts for 10 000 iterations on NWPU VHR-10.v2 and 40 000 iterations on DIOR. For the detector, the learning rate is decayed by a factor of 0.1 at 5000 and 8000 iterations on NWPU VHR-10.v2, and at 24 000 and 32 000 iterations on DIOR. Few-shot fine-tuning lasts for 2000 iterations on NWPU VHR-10.v2 and 10 000 iterations in the other settings. Image patches are randomly cropped to 400×400 pixels for NWPU VHR-10.v2 and 608×608 pixels for DIOR. Multiscale training from 0.5 to 2.0, random flipping, and random rotations of 90°, 180°, and 270° are applied. The batch size is 16 for base training and 8 for fine-tuning.

The teacher network is updated by an exponential moving average (EMA) with momentum $m = 0.999$. ST-FSOD is used as the baseline.

During fine-tuning, the dynamic prototype queue maintains two fixed-length queues of capacity 200 for the projected features and the 1024-D ROI features of each category. The class prototypes $\mathbf{p}_t$ are updated from the queue means using EMA. The student head is optimized by gradient descent, whereas the teacher head is updated at each iteration from the student weights using EMA. For feature synthesis, P-GAN conditions a 64-D noise vector on class prototypes to generate 1024-D ROI features. The discriminator branch uses Adam with a OneCycleLR schedule; this scheduler is separate from the step-decay schedule used for the detector. The P-GAN branch includes a 500-iteration warm-up phase, followed by gradient penalty and gradient clipping. Novel-class features are synthesized every five iterations, and only features whose discriminator confidence exceeds 0.7 are used in the generated-feature classification loss.

TABLE II
MAP (IN %) AT AN IOU THRESHOLD OF 0.5 ON DIOR FOR DIFFERENT METHODS, WHERE THE BASE-NOVEL CLASS SPLIT FOLLOWS [13]. THE BEST AND SECOND-BEST RESULTS ARE MARKED IN BOLDFACE AND UNDERLINED, RESPECTIVELY

Method / Shots	Novel Classes			
	3-shot	5-shot	10-shot	20-shot
OFA [51]	32.8	37.9	40.7	–
FSODM [13]	–	25.0	32.0	36.0
SAM&BFS [52]	–	38.3	47.3	50.9
PAMS-Det [53]	28.0	33.0	38.0	–
CIR-FSD [54]	–	33.0	38.0	43.0
TFACSC [55]	38.0	42.0	47.0	–
SAGS&TFS [55]	–	34.0	37.0	42.0
TSF-RGR [56]	–	42.0	49.0	54.0
ST-FSOD [18]	<u>49.0</u>	<u>49.6</u>	<u>57.3</u>	<u>63.3</u>
Ours	**52.1**	**52.4**	**58.2**	**66.4**

D. Experimental Results and Comparisons

We evaluate the proposed method on DIOR and NWPU VHR-10.v2. Following the established base-novel partitions for each dataset, we use five DIOR splits and one NWPU VHR-10.v2 split. The corresponding results are reported below.

1) Results on DIOR: For the DIOR dataset, we adopt five partitioning strategies. The first split follows the protocol established by Li et al. [13], while the subsequent four splits follow Cheng et al. [14], as detailed in Table I. Across all five splits, we evaluate our FSOD framework under 3, 5, 10, and 20-shot settings.

The experimental results on the DIOR dataset indicate that the proposed method consistently improves detection accuracy across various shot configurations. As shown in Table II, our method achieves mAP scores of 52.1%, 52.4%, 58.2%, and 66.4% under the 3, 5, 10, and 20-shot settings, respectively. These results yield absolute improvements of 3.1%, 2.8%, 0.9%, and 3.1% over prior approaches. Notably, the model demonstrates stable performance in extremely low-shot scenarios, maintaining an mAP above 52% even when trained with only 3 or 5 instances per novel class.

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TABLE III

COMPARISON OF FSOD PERFORMANCE ON DIOR [46] UNDER THE 3-, 5-, 10-, AND 20-SHOT SETTINGS, WHERE THE BASE-NOVEL CLASS SPLIT FOLLOWS [14]. THE BEST AND SECOND-BEST MAP RESULTS ARE MARKED IN BOLDFACE AND UNDERLINED, RESPECTIVELY

Split	Setting	P-CNN [14]	SAGS [55]	SAE-FSDet [20]	G-FSOD [27]	CGK-FSOD [57]	ST-FSOD [18]	**Ours**
split 2	3-shot	18.0	29.3	26.9	27.6	41.6	41.9	**43.1**
	5-shot	22.8	31.6	32.4	30.5	42.0	**45.7**	43.5
	10-shot	27.6	36.4	37.1	37.5	44.7	**50.0**	49.4
	20-shot	29.6	40.2	42.5	39.8	47.5	**53.7**	51.2
split 3	3-shot	14.5	12.6	14.0	14.1	**26.0**	17.7	23.0
	5-shot	14.9	15.5	15.7	15.8	26.0	20.7	**26.2**
	10-shot	18.9	19.7	17.4	20.7	24.8	27.3	**30.6**
	20-shot	22.8	23.8	21.3	22.7	28.7	33.4	**36.0**
split 4	3-shot	16.5	20.9	16.7	16.0	18.0	20.9	**21.9**
	5-shot	18.8	24.8	19.1	23.3	22.6	26.0	**26.4**
	10-shot	23.3	28.6	28.4	26.2	26.7	31.3	**31.6**
	20-shot	28.8	36.1	29.9	32.1	33.1	34.6	**36.7**
split 5	3-shot	15.2	17.5	17.3	16.7	17.9	20.4	**22.0**
	5-shot	17.5	19.7	20.5	21.0	23.8	**25.2**	23.6
	10-shot	18.9	23.7	22.7	25.8	24.8	**33.4**	33.1
	20-shot	25.7	27.7	26.8	31.8	31.3	**38.2**	37.6

TABLE IV

COMPARISON OF FSOD PERFORMANCE ON NWPU VHR-10.V2 [47] UNDER THE 3-, 5-, 10-, AND 20-SHOT SETTINGS. THE BEST AND SECOND-BEST MAP RESULTS ARE MARKED IN BOLDFACE AND UNDERLINED, RESPECTIVELY

Model Name	**3-shot**			**5-shot**			**10-shot**			**20-shot**		
	Base	**Novel**	**All**	**Base**	**Novel**	**All**	**Base**	**Novel**	**All**	**Base**	**Novel**	**All**
Meta-RCNN [9]	87.00	20.51	67.05	87.77	21.77	66.55	87.01	26.98	69.00	87.29	28.24	69.57
FsDetView [58]	87.68	24.56	68.75	87.77	29.55	70.31	87.75	31.77	70.96	87.83	32.73	71.30
P-CNN [14]	82.84	41.80	70.53	82.89	49.17	72.79	83.05	63.29	78.11	83.59	66.83	78.55
TFA [59]	89.35	8.80	65.19	89.60	9.49	64.65	89.95	9.26	65.74	87.26	10.83	65.98
DeRCNN [21]	87.22	37.90	73.33	86.87	46.08	74.63	87.29	62.95	79.99	87.26	64.61	80.46
Retentive R-CNN [60]	90.03	14.46	67.36	90.01	20.29	69.09	90.06	21.99	69.64	90.62	26.57	71.41
G-FSDet [27]	89.11	49.05	77.01	88.37	56.10	78.64	88.40	71.82	83.43	89.73	75.41	85.44
RIFR [61]	94.64	53.17	**82.20**	95.26	62.39	85.40	96.23	72.29	89.04	95.38	79.46	90.61
ST_FSOD [18]	-	60.70	-	-	67.20	-		79.7	-	-	84.8	-
Ours	**96.93**	**61.04**	82.10	**95.98**	**70.69**	**88.40**	**98.20**	**80.33**	**92.80**	**95.83**	**85.56**	**92.80**

Furthermore, Table III reports the results for the remaining four splits. The proposed approach achieves the highest mAP in 9 of the 16 evaluated settings, including all four settings in Split 4. In the remaining settings, it generally remains competitive with the strongest compared methods. These results indicate that the method transfers across different novel-category partitions.

2) Results on NWPU VHR-10.V2: Table IV presents the comparative results under the selected base-novel split. We focus on novel-class performance while also reporting base and overall mAP. The proposed method obtains novel-class mAP values of 80.33% and 85.56% at 10 and 20 shots, respectively. Its base-class mAP remains above 95% across the evaluated settings. At 10 shots, the overall mAP is 3.76 percentage points higher than that of RIFR. These results indicate that the method improves novel-class adaptation while retaining competitive base-class performance.

E. Qualitative Results

We present qualitative results on the DIOR and NWPU VHR-10.v2 datasets in Figs. 3–6.

1) The results in Figs. 3 and 4 illustrate detections of novel-class targets in complex remote sensing scenes under the 20-shot setting. The examples include airplanes, baseball fields, tennis courts, train stations, and windmills, with variations in scale, background, and occlusion. The visual agreement between the detections and ground-truth annotations provides qualitative support for novel-class adaptation.

2) The comparison in Figs. 5 and 6 shows representative cases in which the proposed method detects small targets such as vehicles and windmills that are missed or incorrectly detected by the baseline. The base-class examples in Fig. 6 also show that detections of previously learned categories are retained after fine-tuning. These visualizations complement the quantitative results but do not replace them.

F. Ablation Studies

1) Ablation Study for Each Component: We present the contribution of each component in Table V. The first row is a DABS-only baseline, which adopts a fine-tuning strategy with a frozen backbone and FPN networks and obtains a novel-class mAP of 45.1%. To systematically validate the proposed method, we progressively evaluate the DPLC and P-GAN modules with and without DABS. Adding DABS to the DPLC setting improves the novel mAP from 46.9% to 48.0%, and the full model with DPLC, P-GAN, and DABS achieves the best

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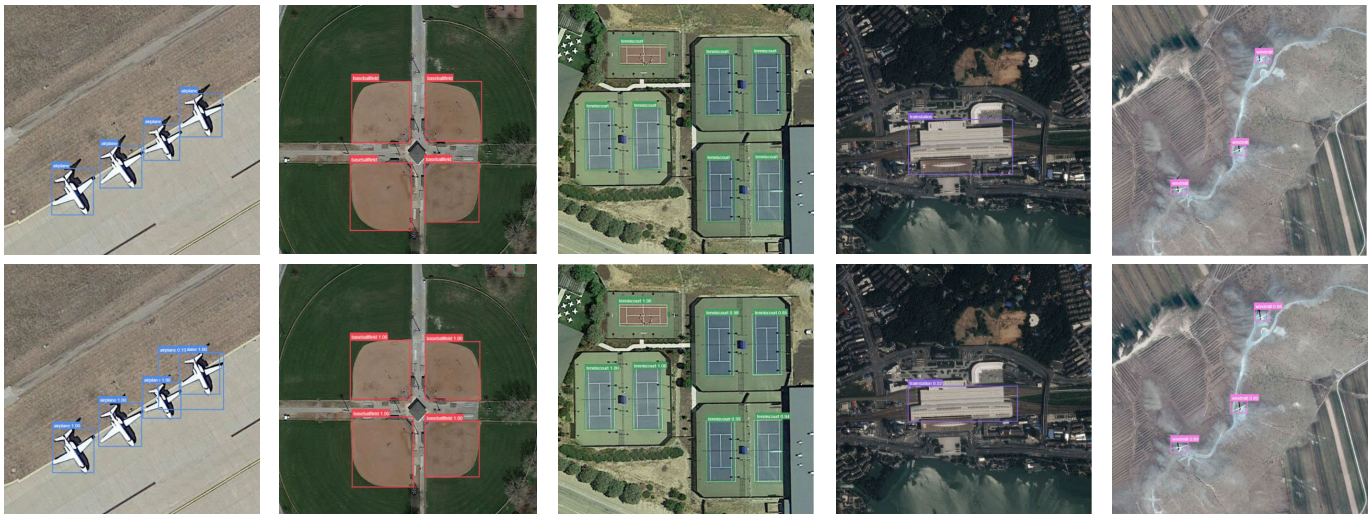


Fig. 3. Qualitative results of the proposed method on DIOR under the 20-shot setting. The first row shows ground-truth boxes, and the second row shows detections for novel classes.



Fig. 4. Qualitative results of the proposed method on NWPU VHR-10.v2 under the 20-shot setting. The first row shows ground-truth boxes, and the second row shows detections for novel classes.

TABLE V

ABLATION STUDY FOR DIFFERENT COMPONENTS ON DIOR. THE FIRST ROW DENOTES THE DABS-ONLY BASELINE, AND THE REMAINING ROWS PROGRESSIVELY EVALUATE DPLC AND P-GAN WITH OR WITHOUT DABS. THE EVALUATION METRIC IS MAP, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	DPLC	P-GAN	DABS	Airplane	Baseball field	Tennis court	Train station	Windmill	Novel	Base	All
DABS-only baseline			✓	49.4	80.0	69.5	11.7	14.4	45.1	70.2	64.0
DPLC	✓			55.1	76.4	71.1	14.0	17.8	46.9	75.3	68.2
DPLC+DABS	✓		✓	56.4	79.5	70.5	17.3	16.5	48.0	74.6	68.0
DPLC+P-GAN	✓	✓		57.7	77.1	71.1	**19.9**	**18.4**	48.8	**75.5**	68.9
Full model	✓	✓	✓	**67.0**	**81.5**	**76.5**	17.4	17.8	**52.1**	75.3	**69.5**

novel mAP of 52.1% and the best overall mAP of 69.5%. This progressive analysis indicates that DABS improves the data-level balance, while DPLC and P-GAN further refine label assignment and enrich the feature space.

2) Ablation Study for DABS: The ablation results suggest that DABS helps mitigate overfitting associated with naive oversampling. As illustrated in Fig. 7, DABS increases the number of detected bounding boxes and improves recall relative to the single augmented balanced sampling strategy under the 3-shot setting. By applying context-aware augmentation variants, such as color jittering, grid occlusion, and small-object duplication, DABS increases training diversity from

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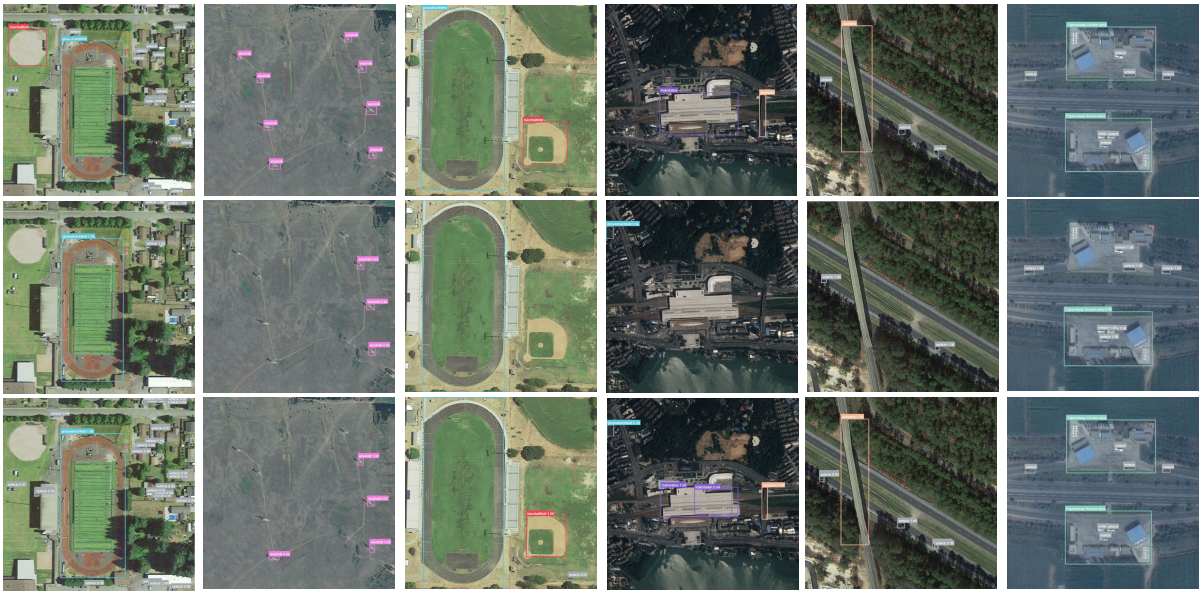


Fig. 5. Qualitative comparison between the baseline and the proposed method on DIOR Split 1 under the 20-shot setting. The first row shows ground-truth boxes, the second row shows baseline detections, and the third row shows detections from the proposed method.



Fig. 6. Qualitative results for base classes on DIOR split 1 under the 20-shot setting.

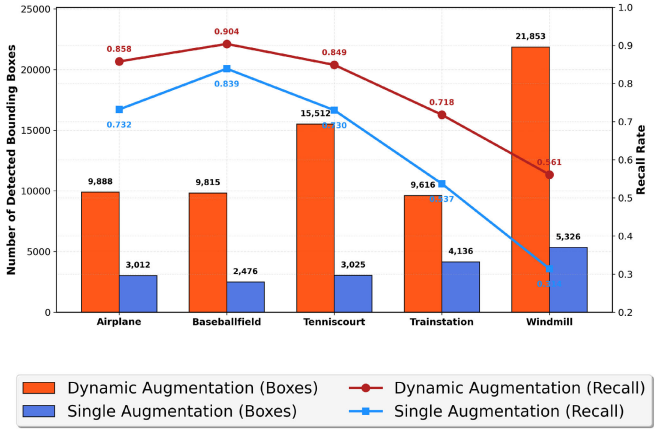


Fig. 7. Number of detected bounding boxes and the recall rate for novel-class objects on the test set are illustrated in the figure. The orange bars and line represent the results obtained using the DABS, while the blue bars and line correspond to those obtained with the single augmented balanced sampling strategy.

limited samples. This can encourage the model to learn more robust feature representations rather than memorizing few-shot idiosyncrasies.

3) Ablation Study for DPLC: To examine the behavior of DPLC, Fig. 8 visualizes the relationship between prototype similarity and the IoU of ambiguous gray-area proposals. In the illustrated examples, proposals with higher prototype similarity tend to have higher IoU with unannotated foreground objects. This observation supports the use of prototype similarity as a signal for attenuating the background loss of potentially mislabeled proposals during few-shot fine-tuning.

4) Ablation Study for P-GAN: To demonstrate how P-GAN's cross-attention mechanism supports semantic alignment, we provide a dual-perspective visual analysis in Fig. 9. First, nearest neighbor proxy retrieval displays the top-3 real image patches retrieved using synthetic ROI features as queries. The retrieval of target category elements indicates that the generated features preserve high-level visual semantics rather than arbitrary distributions. Second, channel-wise activation profiling compares the activation intensities of the top feature channels. The radar charts show that the synthetic features (generated) follow trends similar to those of authentic features (real) and condition vectors (prototype). This consistency suggests that the cross-attention mechanism injects category-specific semantics into the synthesized features and improves semantic alignment.

5) Ablation Study for APCL: We evaluate the impact of the APCL by analyzing its balancing hyperparameter, λ_{APCL}. As shown in Table VI, the best result in this sweep is obtained at $\lambda_{\text{APCL}} = 0.4$. Smaller values provide weaker structural constraints, whereas excessively large values can reduce performance by placing too much emphasis on the auxiliary objective. Furthermore, the t-SNE visualizations in Fig. 10 show that the prototype distributions become more

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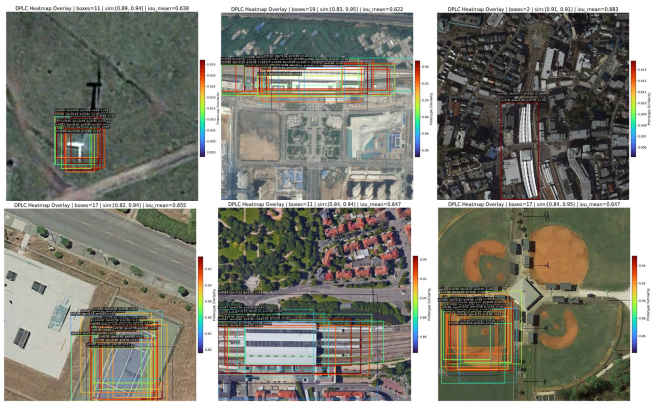


Fig. 8. Correlation between maximum prototype similarity and max IoU for negative proposals during training.

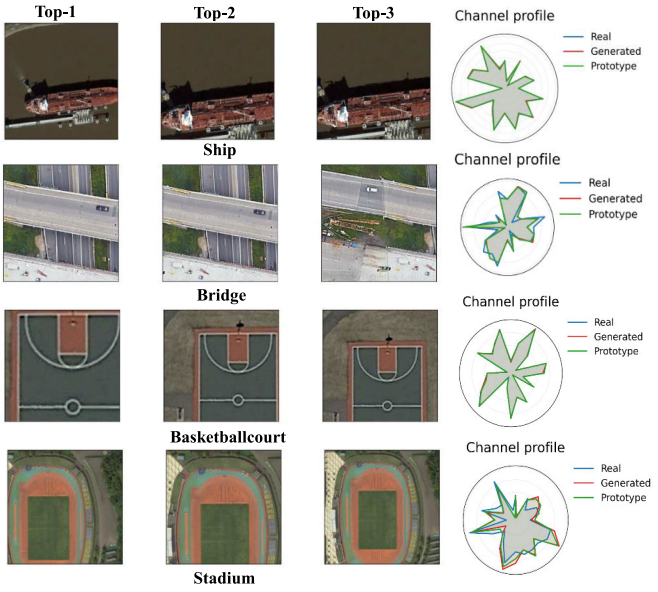


Fig. 9. Semantic alignment of P-GAN features. (Left) Nearest-neighbor real patches retrieved using synthetic features. (Right) Channel-wise activation profiles comparing real, generated, and prototype features.

TABLE VI

ABLATION STUDY FOR THE BALANCING HYPERPARAMETER λ_{APCL} IN $\mathcal{L}_{\text{APCL}}$. THE EVALUATION METRIC IS MAP, AND THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

λ_{APCL}	Airplane	Baseball field	Tennis court	Train station	Windmill	Novel	Base	All
0.2	**86.1**	83.7	81.2	35.2	43.8	66.0	74.3	72.3
0.4	85.7	84.0	81.2	**35.6**	**45.5**	**66.4**	**74.5**	**72.5**
0.6	85.6	**84.5**	**81.3**	34.6	44.1	66.0	74.3	72.3
0.8	84.5	83.3	81.0	35.4	42.5	65.4	74.3	72.1
1.0	84.8	82.2	80.8	34.2	42.4	65.0	74.4	72.0

compact and distinct over training. This evolution is consistent with APCL helping organize the latent feature space for few-shot learning.

6) Ablation Study for ROI Feature Distributions: Fig. 11 visualizes the evolution of ROI feature distributions. In the early training stage, synthesized (red) and real (blue) features are separated. At the later stage, the two distributions become more aligned. This visualization is consistent with the intended role of P-GAN in enriching the novel-class feature space.

TABLE VII

EFFICIENCY COMPARISON

Method	Components		Training Time (s)	mAP (Novel)	Params (M)	GPU-M (MB)	FPS
	DPLC	P-GAN					
ST-FSOD [18]	-	-	0.479	49.6	75.62	613.23	9.16
Ours	✓		0.296	46.9	97.75	696.15	8.83
		✓	0.465	46.4	97.75	696.15	8.83
	✓	✓	0.528	52.4	97.75	696.15	8.83

G. Computational Efficiency

To comprehensively assess the practical cost-performance tradeoff and the incremental computational demands introduced by our proposed framework, we conducted an efficiency evaluation. We recorded the training time measured in seconds per iteration during the fine-tuning stage and the inference-stage metrics, including parameter count (Params), GPU memory footprint (GPU-M), and frames per second (FPS). The outcomes, compared against the baseline ST-FSOD, are illustrated in Table VII.

During inference, neither the dynamic prototype queue used by DPLC nor the P-GAN feature-synthesis process is executed. The parameter counts in Table VII are reported for the complete instantiated model and therefore include the training-time modules; they should not be interpreted as the parameters actively used to generate detections. Across the ablated variants, the measured GPU memory and FPS are identical (696.15 MB and 8.83 FPS), indicating that DPLC and P-GAN do not add runtime cost among these variants under the reported measurement protocol.

During the training phase, the integration of both DPLC and P-GAN introduces a slight computational load, increasing the per-iteration training time to 0.528 s compared to 0.479 s for the baseline. However, since the few-shot fine-tuning phase typically involves fewer iterations than the base training phase, this extra training cost remains within an acceptable range. While our overall model exhibits a marginal increase in parameter count and a slight drop in FPS compared to ST-FSOD, the observed performance gain suggests a favorable efficiency tradeoff.

V. CONCLUSION

In this work, we address the critical challenges of FSOD in remote sensing imagery—particularly severe data scarcity for novel classes and the prevalence of incomplete annotations that lead to false-negative training samples. To this end, we propose a P-GAN framework that integrates class-aware prototype learning with adversarial feature generation to enrich semantic representations of novel objects from limited support examples. Furthermore, we introduce a DPLC within a teacher–student architecture to identify and reweight ambiguous negative proposals, thereby mitigating the adverse effects of missing annotations.

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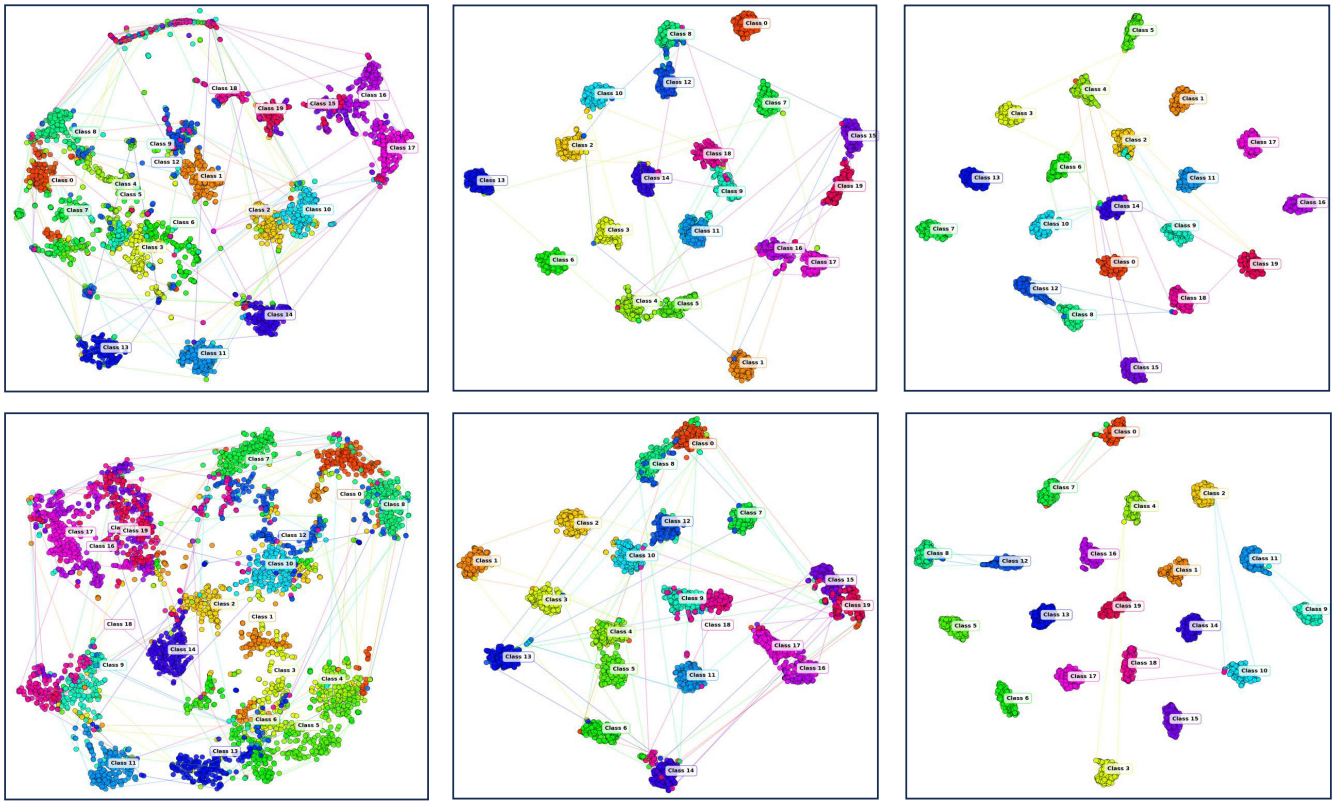


Fig. 10. t-SNE visualization results reflect the distribution of class prototypes in the current latent space. The first row presents the prototype feature distribution of the DIOR dataset under split 1, while the second row shows the prototype feature distribution under split 2. From left to right, the figures represent the initial state, early training stage, and end-of-training stage outcomes, respectively.

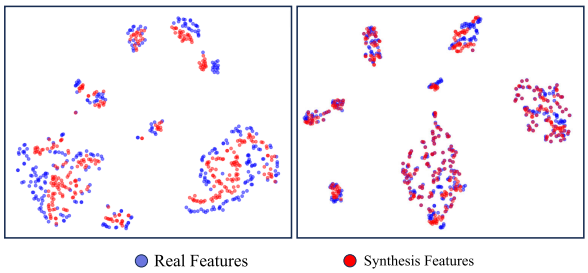


Fig. 11. Comparison of synthesized and real ROI feature distributions during the early and post-training phases of P-GAN.

Experiments on the benchmark datasets show that the proposed method improves performance over the compared FSOD approaches in several shot settings while addressing the influence of incomplete annotations. Limitations remain, particularly in handling extreme intraclass variation and cross-domain generalization under ultralow-shot regimes. Future work will explore self-supervised pretraining with unlabeled remote sensing data and integrate explicit uncertainty modeling to further improve reliability and adaptability in real-world deployment scenarios.

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